25901093 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBCH200 - BIOCHEMICAL TESTS IN DISEASE DIAGNOSIS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice questions

Answer all questions

Each question carries **1** marks

1. Which ethical responsibilities uses external quality assessment: [U] / [CO1]

   a). Quality Assurance and Ethical Practices

   b). Analytical stage

   c). Role of codes

   d). Ethics in communication

2. The normal 24-hour urinary uric acid excretion is [An] / [CO3]

   a). 50–150 mg/day

   b). 100–300 mg/day

   c). 250–750 mg/day

   d). 800–1200 mg/day

3. Which test reflects long-term blood glucose control in diabetes? [K] / [CO4]

   a). FBS

   b). HbA1c

   c). RBS

   d). PPBS

4. Which of the following is not a component of a routine blood test? [U] / [CO1]

a). Hemoglobin b). RBC count

c). Urea d). WBC count

5. Elevated bilirubin levels generally indicate: [U] / [CO3]

a). Kidney dysfunction b). Lung disease

c). Liver dysfunction d). Cardiac disease

[1x5 = 5]

Part B

Fill in the Blanks questions

Answer all questions

Each question carries **2** marks

6. Blood specimens must be transported in _____ containers to prevent leakage [U] / [CO1]

7. Type 2 diabetes mellitus is mainly due to _____ resistance. [K] / [CO2]

8. A device that uses a biological component linked to a transducer to convert biochemical reactions into measurable signals is_____ [K] / [CO4]

9. The first morning urine sample is preferred fot protein and chorionic gonodotropin measurement because it is more _____. [U] / [CO1]

10. The normal albumin–globulin (A/G) ratio ranges from _____ to _____ ng/ml. [E] / [CO3]

[2x5 = 10]

Part C

Short Answer Type Questions

Answer any **5** questions

Each question carries **3** marks

11. Define analytical factors in clinical laboratory testing. [K] / [CO1]

12. Label the main advantage of HbA1c testing. [K] / [CO2]

13. A patient presents with itching and wheezing. DC shows eosinophils = 9%. Analyse the condition. [An] / [CO4]

14. Describe the normal volume and colour of CSF and explain their clinical relevance. [U] / [CO1]

15. Analyze the changes in AFP levels from fetal life to adulthood and their diagnostic importance. [An] / [CO3]

16. Analyse the reason why serum creatinine is considered a reliable indicator of GFR. [An] / [CO3]

17. Explain internal quality control. [U] / [CO1]

[3x5 = 15]

Part D

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

18. Evaluate the clinical significance of Triglyceride. [E] / [CO1]

19. Describe the Cyanmethemoglobin (Drabkin's) method for estimation of hemoglobin. [U] / [CO1]

20. Describe the role of the Glucose Challenge Test in screening for gestational diabetes mellitus. [U] / [CO2]

21. Summarize the physiological principles underlying renal function tests and state their clinical applications. [U] / [CO3]

22. Explain how robotic pipetting improve laboratory outcomes. [U] / [CO4]

23. Outline next-generation sequencing (NGS). [An] / [CO4]

[5x4 = 20]